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Dissertation

FROM BLACK BOXES TO GLOBAL INTERPRETABILITY: DISTILLING DEEP NEURAL NETWORKS INTO DIFFERENTIABLE TREE MODELS

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Abstract

Predictive models deployed in high-risk fields such as medical care must be both accurate and transparent to be trusted. While deep learning models usually achieve the best performances, they are opaque in terms of explainability. This thesis implements the Tree-Network-Tree architecture that takes the insights from such complex networks and transfers them to inherently interpretable tree-based models by performing knowledge distillation. The student model is a Gradient Boosting Decision Tree architecture that uses Soft Decision Trees as weak learners, making a fully differentiable ensemble that preserves the advantages of both neural networks and trees, plus the performance of the teacher network. It learns via backpropagation like the neural networks, but because the raw data passes without any alterations through every single node, opposed to NNs, which abstract the input into complex representations at each hidden layer, every part of the network remains transparent and explainable.

The models were tested on the task of Intensive Care Unit mortality risk prediction on the MIMIC-III dataset. In order to handle the medical data sparsity, the pipeline integrates a masking mechanism into the Soft Trees that adds learnable parameters for the missing values. This way, the models see absence of data as extra features and use them in the decision-making process.

Results show that the student network did, in fact, preserve the teacher's performance. Visualizing the core of the network and the patients' probabilistic trajectory across the trees revealed a lot of insights about how the model uses the features and even the absence of them to draw high and low risk clinical profiles and splits the patients according to them.

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Chapter 1

Introduction

In sensitive environments such as intensive care medicine, the trustworthiness of a predictive model is just as important as its raw accuracy. While modern deep learning architectures excel at detecting complex patterns in data using multiple layers of abstraction, their underlying decision-making processes are opaque. Because these networks rely on millions of learned parameters and alter inputs through non-linear, unreadable transformations, it is generally infeasible to trace how a specific prediction was made. This inherent "black-box" nature has restricted their adoption in fields where decisions come with great responsibility. The focus in machine learning is increasingly moving to models that can explicitly reveal their reasoning while not sacrificing their predictive power. Traditionally, the pursuit of transparency points toward standalone decision trees [1]. To handle their high variance and effectively manage the sparse, missing data common in clinical datasets, practitioners rely on ensemble methods like Gradient Boosting Decision Trees (GBDTs) [4]. While these hard-split ensembles provide excellent predictive robustness, their massive scale usually transforms them into not so transparent "gray boxes", sacrificing some of their interpretability. Furthermore, GBDTs don't have the end-to-end differentiable learning capabilities that make deep neural networks so exceptionally powerful.

Trying to bring together both power and interpretability, this thesis explores and implements the Tree-Network-Tree (TNT) learning framework [12]. The TNT architecture is designed to transfer knowledge from deep neural networks to robust tree structures, keeping the advantages of both paradigms. In this pipeline, an initial tree ensemble processes the raw input data to generate dense knowledge embeddings. These embeddings are then fed into a DNN, which acts as a strong, high-performing "teacher" model. Finally, the "soft targets" produced by the teacher network are distilled into a transparent student model: a distillable Gradient Boosting Decision Tree (dGBDT) composed of differentiable Soft Decision Trees (SDTs).

In the context of this thesis, I focus heavily on the inherent interpretability and explainability of the TNT framework, specifically adapting the architecture to process the

complex, irregular time series typically found in critical care monitoring. I used the MIMIC-III database to evaluate the framework on the task of Intensive Care Unit (ICU) mortality risk prediction. Because clinical temporal data is notoriously noisy, with irregular sampling intervals, I implemented a masking mechanism for the soft trees. Instead of relying purely on mean imputation, this approach treats missingness as a distinct feature, similar to the default routing directions that algorithms like XGBoost learn for handling missing data.

By examining both global and local interpretability, this work shows that the distilled tree models manage to keep predictive performance and robustness, while also providing human-readable clinical insights. Looking at the learned weights of internal nodes, the probabilistic patient trajectories across trees, and the impact of missing features, I show how a deep network's complex reasoning can be successfully translated into powerful, transparent models required for high-risk medical settings.

1.1 Contributions

This thesis presents my approach to the TNT architecture as follows:

I implemented the dGBDT algorithm presented by Li et al. [12] in their paper, integrating the sequential boosting paradigm and the distillation loss over the base SDT module. Additionally, I built a MLP and the XGBoost embeddings mechanism and linked all modules together to test the framework on the UCI Wisconsin breast cancer dataset.

In order to make the student models work directly on the raw data without imputing the missing values, I adapted the SDT by adding a missingness mask with learnable parameters, making the model use the absence of data as a feature.

The main challenge was moving to the ICU mortality risk prediction task, because processing temporal data required a set of modifications:

- Each sample was split in static hourly snapshots and embedded separately to keep the temporal dimension of data intact
- The selected teacher model, the ConvNet from the ISeeU paper, was adapted to accept the embeddings as input
- The input was transformed, as the student models require unidimensional vector samples. For this, I adopted two separate methods and tested both: flattening the input from 22×48 to a unidimensional 1056 feature vector, and aggregating the temporal data in 4 representative statistics (min, max, mean, variance).

Moving forward, I focused on creating visual representations of trees, nodes, feature importances and patient trajectories across the trees. I showed how looking inside these transparent models can reveal unexpected insights about the model's decision-making process, highlighting the inherent interpretability of the tree models.

Chapter 2

Related Work

2.1 Deep Models

Deep Neural Networks are a subclass of machine learning algorithms modeled after the mammalian brain, processing the input information in different layers of abstraction. As the data flows through the network, low level features are combined into higher level features, allowing for better expression of features and ability to map complex patterns[16]. Deep models are constantly achieving state-of-the-art performance in most machine learning applications and are used in a lot of real world applications at this moment. However, researchers have increasingly prioritized other quality factors other than performance accuracy of the model, such as robustness to data distribution changes and perturbations, safety and security aspects and the ability of revealing the decision process in order to prove its trustworthiness to a human user [15]. Because DNNs rely on millions of learned parameters and apply complex, non-linear transformations that alter input shapes and values layer by layer, it is generally infeasible to interpret intermediate representations and try to understand how the decisions are made when making predictions. Because of this, deep networks have been labeled as "black boxes". This opacity has restricted their adoption in sensitive fields such as medicine [14, 8].

2.2 Decision Trees and Boosting

Decision Trees represent a class of machine learning models that are structured in a tree-shape, where at each node the input is routed to a different branch based on some rules the model learns, until it reaches a leaf node that has an assigned value, representing the prediction of the tree. The most widely known algorithm for training decision trees is CART (Classification and Regression Trees). It was described for the first time in 1984 by Breiman et al. in their book [1] and is still used today as a standalone model or in ensembles. The algorithm starts with the whole set of data at the root node and searches

for an optimal split that leads to the "purest" possible resulting segments. The process continues recursively in every segment until stopping criteria are met, creating a binary tree that has a "true or false" rule at every node.

While individual decision trees are highly interpretable, they are often prone to high variance and overfitting. Because of this, ensemble techniques such as boosting were created. Originally introduced by Freund and Schapire [6], boosting operates by sequentially building multiple "weak" learners, usually decision trees, where each new tree focuses specifically on minimizing the residual error made by the previous one. Building upon this theoretical principle, the gradient boosting framework was dramatically advanced in 2016 when Chen and Guestrin introduced XGBoost (eXtreme Gradient Boosting) [4]. XGBoost revolutionized tree-based methods by providing a highly scalable architecture that incorporates L1 and L2 regularization to penalize model complexity, alongside sparsity-aware split finding and parallelized tree formation.

2.3 Soft Trees

Soft Trees introduce a probabilistic approach to the traditional routing mechanism of decision models. Unlike standard trees where an input takes a single hard path, in a soft tree, the data flows across all branches simultaneously with different probabilities. A widely adopted architecture is the Soft Decision Tree (SDT) developed by Nicholas Frosst and Geoffrey Hinton[9]. The routing process starts with the input at the root node holding 100% of its mass. At each internal node, the sum of the feature contributions, alongside a bias term, is evaluated using a sigmoid activation function. This produces a continuous value in [0, 1] interval, which dictates the exact probability of routing the data down the left child branch versus the right child branch. As the data traverses the network, its initial mass is continuously multiplied by these left and right routing probabilities at every split, ultimately resulting in a final mass contribution distributed across the leaves. Because of their differentiable architecture, soft trees can be trained using gradient descent methods and used as standalone models or linked together into powerful, interpretable ensembles.

2.3.1 Knowledge Distillation

While deep neural networks usually obtain the strongest predictive performance, their computational cost and opaque, "black-box" nature limit their deployment in sensitive environments like healthcare. To address this problem, Knowledge Distillation (KD) was introduced. The core concept, initially explored under the umbrella of model compression by Buciluǎ et al. [2], was formally defined by Hinton in 2015 [10].

In the standard knowledge distillation framework, a smaller, simpler "student" model

is trained to mimic a larger, highly complex "teacher" model. Instead of training the student strictly on the hard, binary ground-truth labels of the dataset, it is trained to mimic the "soft targets" or output probabilities produced by the teacher. These soft labels contain what Hinton describes as "dark knowledge", the relative predicted probabilities that reveal how the teacher model generalizes and relates different classes one to another. By copying the teacher's output distribution, the student converges faster and achieves a higher performance than it would if trained on the raw data alone.

2.3.2 Distillable Gradient Boosting Decision Trees

In gradient boosting, an ensemble of trees is sequentially trained to approximate the accumulated residual errors of prior iterations [7]. Starting from this base, recent advancements incorporated differentiability directly into tree architectures. Traditional Gradient Boosting Decision Trees (GBDTs) use strict, discrete splits, that do not allow the use of end-to-end gradient optimization. To address this limitation, Soft Decision Trees (SDTs) replace hard routing rules with a probability-calculated sigmoid gating function at each internal node [9]. Because the input data routes through all branches simultaneously with varying probabilities, SDTs are fully differentiable and can be trained via standard backpropagation.

Integrating SDTs as base learners leads to the formulation of soft Gradient Boosting Machines (sGBM) [5], which can optimize multiple trees simultaneously using gradient descent. However, to fully transfer the complex, multi-scale feature representations learned by a deep neural network (DNN) into an interpretable tree structure, a specialized distillation mechanism is required. This forms the theoretical basis of the distillable Gradient Boosted Decision Tree (dGBDT) architecture.

Unlike standard boosting, which targets the raw ground-truth error, dGBDT is trained with a composite distillation loss that mimics the "soft labels" generated by a DNN teacher. The architecture is composed of M sequential SDTs ($\{h_m\}_{m=1}^{M}$), characterized by their parameters θ_m and output o_m. For a given sample x^i with a true label y^i and a soft label t^i produced by the teacher network, the distillation loss utilized to calculate the residual for the m-th tree is defined as:

$$dl(s^i_{m-1}, \alpha, y^i, t^i) = \alpha T^2 ||s^i_{m-1} - t^i||^2 + (1-\alpha)||s^i_{m-1} - y^i||^2 \tag{2.1}$$

In this formulation, $s^i_{m-1} = \sum_{j=0}^{m-1} o^i_j(x^i, \theta_j)$ represents the cumulative output of the ensemble up to the previous tree, bridging the predictions made so far. The parameter α is a trade-off weighting coefficient that balances the model's reliance on the teacher's knowledge against the absolute ground truth, while T represents the distillation temperature used to soften the teacher's target distribution [12].

During the sequential training process, the residual r^i_m that the m-th tree must esti-

mate is derived directly from the negative gradient of this distillation loss with respect to the cumulative prediction s^i_{m-1}. The global loss $\mathcal{L}$ for the entire dGBDT ensemble is then minimized by summing the Mean Squared Error across all individual soft trees:

$$\mathcal{L} = \sum_{m=1}^{M} ||r_m - o_m||_2^2$$

By optimizing this loss function, the dGBDT explicitly captures the "dark knowledge" embedded within the DNN teacher. Because the data flows through the ensemble in a loop-free directed acyclic graph way, the final model preserves the inherent transparency of decision trees, allowing individual predictions to be explicitly traced and explained based on feature contributions.

2.4 ConvNet

Convolutional Neural Networks (CNNs), originally designed for image processing, have proven to be highly effective at capturing temporal dependencies and trends in time-series data. In the context of Intensive Care Unit (ICU) mortality prediction, patient monitoring generates temporal series of laboratory results and vital signs that are noisy and sampled at different rates.

To process this complex temporal data from the MIMIC-III database, Caicedo and Gutierrez introduced an interesting Convolutional Neural Network (ConvNet) in the ISeeU paper [3]. The model utilizes a multibranch convolutional architecture designed to capture clinical trends across different time scales within a patient's first 48 hours of ICU admission. The main part of the network consists of three parallel Conv2D branches, each configured with distinct temporal filters of 3, 6, and 12 hours. The feature maps extracted from these branches are pooled, concatenated, and fed into dense layers. The original methodology relies on a specific preprocessing and imputation method, utilizing forward and backward filling combined with mean imputation to handle the missing values from the clinical sequences. Within the scope of this thesis, this powerful multibranch ConvNet serves as the high-performing "teacher" model, that is distilled into the dGBDT student.

Chapter 3

Experiments

3.1 Implementation

I used the PyTorch framework to implement the Distillable Gradient Boosting Decision Trees architecture and a Multilayer Perceptron that serves as the N in the T-N-T framework for the Proof of Concept part of this paper. For the rest of the paper, the ISeeU ConvNet is used.

For the tree embeddings I used XGBoost. After training the tree ensemble on the current fold, each sample is passed through and each tree outputs a leaf index. All leaf indices are sent to the DNN, which takes the indices as input, concatenates them in groups and uses one Embedding layer per group to reduce the dimensionality of the input. The rest of the DNN is kept intact as just the input dimension is modified.

For the Soft Decision Tree I adapted a public implementation from GitHub[1] of the Nicholas Frosst and Geoffrey Hinton SDT to my needs. More specifically, I solved `NaN` handling by adding a mask for the missing values and imputed them as 0. This way, when a value is missing, its weight is ignored because the feature value is 0, but the mask value is 1, so its weight gets activated, so the model learns a default direction when the feature is missing, simulating the default direction that XGBoost learns for handling sparse data. The raw features and the mask are concatenated, doubling the input feature dimension ($x_{\text{augmented}} = [x_{\text{zero_filled}} \,;\, m_{\text{mask}}]$)

The Distillable Gradient Boosting Decision Tree is an ensemble of Soft Trees that are linked together in a boosting manner, meaning that each tree is minimizing the residual error from the previous tree. As we iterate through the trees we store what is the sum of the previous trees so far to compute the residual that the current tree needs to estimate. To enforce the sequential boosting paradigm, the cumulative prediction s_{m-1} is detached from PyTorch's autograd graph. This prevents backpropagation from modifying the parameters of previously trained trees, ensuring each successive tree is strictly fitted to correct the

[1] https://github.com/xuyxu/Soft-Decision-Tree

residual errors of its predecessors. The training is done using a Stratified 5-fold split on the dataset, with early stopping and saving the best model for each fold. Then the saved model is loaded and used as a teacher for the dGBDT, using the distillation loss described by Equation 2.1. The residual of each tree is composed of:

- residual relative to the ground truth
- residual relative to the teacher prediction

each one weighted by the factor α and $1 - \alpha$ respectively.

3.1.1 Evaluation Metrics

The two performance metrics I used for evaluation were: AUROC (Area Under the Receiver Operating Characteristic curve) and AUPRC (Area Under the Precision-Recall Curve). Rather than evaluating the model based on a single threshold point, both metrics assess performance across a continuous range of classification thresholds. The values are in the [0, 1] interval, with 1 being the best possible result.

AUROC measures the model's overall ability to separate the two classes (e.g., benign vs malignant tumors). It is calculated by plotting the True Positive Rate (TPR / Recall) against the False Positive Rate (FPR) as the classification threshold t varies from 0 to 1.

The formulas for these rates are:

$$\text{TPR}(t) = \text{Recall}(t) = \frac{\text{TP}(t)}{\text{TP}(t) + \text{FN}(t)} \tag{3.1}$$

$$\text{FPR}(t) = \frac{\text{FP}(t)}{\text{FP}(t) + \text{TN}(t)} \tag{3.2}$$

where TP, TN, FP and FN represent True Positive, True Negative, False Positive, and False Negative, respectively.

The AUROC score represents the total area under this curve, calculated mathematically as an integral with respect to the False Positive Rate (FPR):

$$\text{AUROC} = \int_0^1 \text{TPR}(FPR)\, dFPR \tag{3.3}$$

While AUROC is a standard metric, it can present an overly optimistic view of performance in highly imbalanced datasets (e.g., when benign cases significantly outnumber malignant ones). To account for this, I also used AUPRC, which plots Precision against Recall across the thresholds' interval.

Precision quantifies the fraction of the model's positive classifications that are correct,

while Recall represents the fraction of positive samples successfully identified by the model:

$$\text{Precision}(t) = \frac{\text{TP}(t)}{\text{TP}(t) + \text{FP}(t)} \tag{3.4}$$

The continuous formula for the area under the Precision-Recall curve integrates Precision over the change in Recall (R):

$$\text{AUPRC} = \int_0^1 \text{Precision}(R)\, dR \tag{3.5}$$

3.1.2 Proof of Concept

First experiment was conducted using the **UCI Wisconsin breast cancer** dataset, where the columns are 30 numerical features that describe the characteristics of a tumor and the label is the binary diagnosis, malign or benign. The features describe the following 10 measurements in three dimensions: *radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension.* The dataset has only 569 samples (357 benign and 212 malign) and has no missing values, so it is really easy to use and fast to train models over it. It was chosen just to test the architecture and prove that it works.

Tree models are more than capable of handling missing data so variations of the experiment where random features were removed from samples were tested. First 20% of the values were removed and then 40%, to see how the model performs with sparse data.

For each sparsity level I tested multiple tree-based, DNN and Tree-NN fusions to see if the distillation has any real advantages over the normal models. For this dataset, the results were not very conclusive because most models performed really well. In Table 3.1 we can see that all models that used some kind of tree ensemble achieved AUROC and AUPRC close to 0.99.

On the fully complete dataset (0% sparse), almost all models achieved high predictive performance, with AUROC and AUPRC values close to 0.99. However, as sparsity increased, standard neural networks (DNN) and single Soft Decision Trees (SDT) experienced a significant drop in performance. At 40% sparsity, the SDT's AUROC dropped to 0.8188, and the DNN's AUROC fell to 0.8324.

In contrast, the models utilizing a Tree-Network fusion with the default routing for sparse data obtained exceptional performance. The GBDT-DNN teacher model maintained an AUROC of 0.9811 with 40% sparsity by leveraging XGBoost's structural embeddings. Most notably, the distilled dGBDT student models (DNN-dGBDT and GBDT-DNN-dGBDT) showed great results, achieving AUROC scores of 0.9921 and 0.9910 respectively, and AUPRC scores above 0.9870 under 40% sparsity. These numbers show that the proposed distillation method successfully transfers the robust reasoning of GB-

DTs to interpretable soft tree ensembles without altering predictive accuracy.

It is also worth mentioning that the default, publicly available implementation of the SDT without the mask for missing values was first used for testing, and the `NaN` values were imputed with the mean of the train split for each feature. The results were a lot worse than the ones in the current setting, with AUROC and AUPRC scores under 0.9. This happened because the teacher was trained on sparse data and learned how to use the missing values, while the student saw the mean value of the feature and learned from the teacher what to do with the mean, instead of learning what to do with missing data.

Methods		Cancer(0% sparse)		Cancer(20% sparse)		Cancer(40% sparse)	
		AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
T	GBDT	0.9791	0.9667	0.9741	0.9611	0.9762	0.9712
	SDT	0.9675	0.9659	0.9071	0.9024	0.8188	0.7980
N	DNN	0.9587	0.9503	0.8737	0.8131	0.8324	0.7828
T & N	GBDT-DNN	0.9893	0.9846	0.9856	0.9813	0.9811	0.9755
	DNN-dGBDT	0.9955	0.9943	0.9910	0.9886	0.9921	0.9889
TNT	GBDT-DNN-SDT	0.9927	0.9909	0.9920	0.9789	0.9900	0.9857
	GBDT-DNN-dGBDT	0.9949	0.9939	0.9912	0.9884	0.9910	0.9874

Table 3.1: Results for the UCI Wisconsin breast cancer dataset with different models

3.1.3 Validating on a complex dataset

Moving further, I used the **MIMIC-III**[11] database that contains clinical information about adult patients and their hospital admissions, such as demographics, vital signs, laboratory measurements, survival data and other medical data. This database is very popular because it is the first publicly available option for such a huge amount of clinical data. It consists of over 50 thousand admissions to critical care units for 38 thousand distinct patients. It became the benchmark for machine learning in healthcare, for the task of predicting the Intensive Care Unit mortality risk. Because MIMIC-III reflects real and complex medical environments, the data is very noisy and often contains missing values. In critical care, the interpretability of a decision is just as vital as its accuracy. MIMIC-III provides a rigorous environment to demonstrate how the TNT framework can leverage tree models to handle low-quality input data, utilize deep neural networks (DNNs) for high predictive performance, and ultimately use knowledge distillation to provide the transparent, explainable decision paths required in high-risk medical settings.

Cohort Selection

To ensure a robust and reproducible approach, the data preprocessing and imputation pipeline for this study uses the methodology established by Caicedo-Torres and Gutierrez[3]. Data was filtered to only include patients that are over 16 years old at the time of admission, with the length of stay exceeding 48 hours. To maintain the statistical independence of the dataset instances, only the first ICU admission was considered for patients with multiple admissions.

Feature Extraction

For each patient, a comprehensive set of 22 clinical variables was extracted spanning the first 48 hours of their ICU stay. These features mostly correspond to the parameters utilized in the Simplified Acute Physiology Score II (SAPS-II), with several key modifications to optimize the data for machine learning.

- Categorical SAPS-II concepts, such as the "type of admission" and "chronic disease," were decomposed into individual binary indicator variables (e.g., surgical admission, elective admission, presence of AIDS, metastatic cancer).
- The Glasgow Coma Scale (GCS) was separated into its three distinct physiological components (eyes, motor, and verbal) rather than using a single aggregated score.
- Measurements for PO_2 and FiO_2 were included as distinct, separate inputs rather than using their quotient.
- Diastolic blood pressure was also added to the feature set to provide a more comprehensive physiological profile.

The final list of 22 features:

- **Static and Demographic Features:** Baseline risk profile upon admission.
  - *Age:* Patient's age.
  - *Elective admission:* Indicator specifying if the ICU stay was pre-planned.
  - *Surgical admission:* Distinguishes post-surgical recovery from medical emergencies.
  - *AIDS:* Chronic comorbidity representing severe immune system compromise.
  - *Metastatic cancer:* Chronic comorbidity denoting advanced malignancy.
  - *Lymphoma:* Hematological malignancy altering the immune baseline.

- **Temporal Features:** High-frequency vitals and labs tracking physiological stability.
  - *Heart rate:* Cardiovascular metric indicating systemic stress or instability.
  - *Systolic BP:* Maximum arterial pressure during heart contraction.
  - *Diastolic BP:* Minimum arterial pressure when heart rests between beats.
  - *Temperature:* Vital parameter for detecting fever or severe hypothermia.
  - *Urine output:* Critical indicator of systemic organ perfusion and renal function.
  - *GCS (Eyes, Motor, Verbal):* Glasgow Coma Scale values.
  - *WBC:* White blood cell count.
  - *BUN:* Blood Urea Nitrogen, test for assessing kidney function.
  - *Bilirubin:* Evaluates liver function and identifies potential organ damage.
  - *Bicarbonate:* Indicates the patient's metabolic state and acid-base balance.
  - *Potassium:* Essential for maintaining proper cardiac electrical function.
  - *Sodium:* Key for regulating systemic fluid balance.
  - FiO_2*:* Fraction of inspired oxygen (external source).
  - PO_2*:* Pressure of oxygen (in blood).

Because clinical data is notoriously noisy, extensive preprocessing was required. Disparate concepts (such as temperatures recorded in both Celsius and Fahrenheit) were standardized, and clear unphysiological values (such as negative blood pressures or heart rates ≥ 300 bpm) were discarded as artifacts. Furthermore, due to MIMIC-III's privacy protocols which artificially shift the ages of patients over 89 to 300 years old, all patient ages greater than 80 were clipped to exactly 80 years.

The temporal nature of ICU monitoring means that variables are sampled at vastly different rates, resulting in highly irregular time series with significant missing observations. To address this, the data was structured into a 22×48 observation matrix for each patient. Static features, such as admission type and age, were broadcasted across all 48 hourly timestamps. For temporal variables containing multiple measurements within a single hour, the values were averaged, except for urine output, which was summed. To resolve missing hourly data points, a combination of forward and backward filling imputation was applied to the time series. In instances where a patient was missing all FiO_2 values, they were imputed to a standard, normal baseline value of 0.2. Finally, if a patient did not have a single recorded observation for a specific variable across their entire stay, the entire physiological time series for that feature was imputed using the overall empirical mean of the cohort.

TNT Adaptation for Temporal Sequences

Applying the Tree Network Tree architecture to temporal sequences requires a series of modifications in order to make each component accept the new input dimension and try to understand as much as possible from it.

Stage 1: Tree Embeddings (XGBoost) Because GBDTs cannot natively process raw 3D sequence tensors ($N \times 22 \times 48$), I reformatted the patient sequences into independent hourly snapshots turning the whole dataset into a 22 feature by almost 500 thousand entries set. I trained an XGBoost model containing 60 decision trees on these snapshots. Once trained, the GBDT was evaluated in leaf-prediction mode, mapping each temporal state to a vector of leaf indices. This translated the input space from 22 features to 60, while preserving the temporal dimension of the data.

Stage 2: ConvNet Teacher (ISeeU) To capture multi-scale clinical trends, this representation is processed by a multibranch convolutional architecture (modeled on the ISeeU ConvNet). The input is passed through three parallel Conv2D branches with temporal filters of 3, 6, and 12 hours. The outputs of these branches are pooled, concatenated, and fed into fully connected layers with dropout (0.45) and batch normalization. The final output is passed through a sigmoid layer to predict ICU mortality. This network is trained using class weighted binary cross-entropy to overcome the class imbalance (9.6% mortality rate), and its outputs serve as the soft targets for distillation.

I made a modification in the original ConvNet to accept the new input size and pass every single hour vector of leaf indices through the same Embeddings layer. This ensures that the convolutional filters can be still applied to the temporal axis of the input.

Stage 3: dGBDT Student The student model is a distillable GBDT (dGBDT) composed of $M = 6$ Soft Decision Trees of depth 3, operating on a flattened, 1056-dimensional feature vector (22×48). The modifications are also illustrated in Figure 3.1.

Again I measured the AUROC and AUPRC for a set of different experiments, in order to test the performance of various architectures based on trees, neural networks or fusion of both. The results are presented in Table 3.2. Looking first at the standalone models, the neural network (Conv) achieved an AUROC of 0.8484 and an AUPRC of 0.4312. This is the ConvNet from the ISeeU paper[3] without any adaptation. This network outperformed the tree models, but the SDT achieved competitive results, considering it is a single tree model. The standalone dGBDT performed worse than the CART baseline. Moving to the Tree-Network fusions, the GBDT-Conv model(a "T-N" pattern) demonstrated a significant performance leap, achieving an AUROC of 0.8739 and an AUPRC of 0.4796. This improvement validates the theory that using a robust tree ensemble like GBDT at the input stage effectively handles tabular data by creating structured embeddings for the

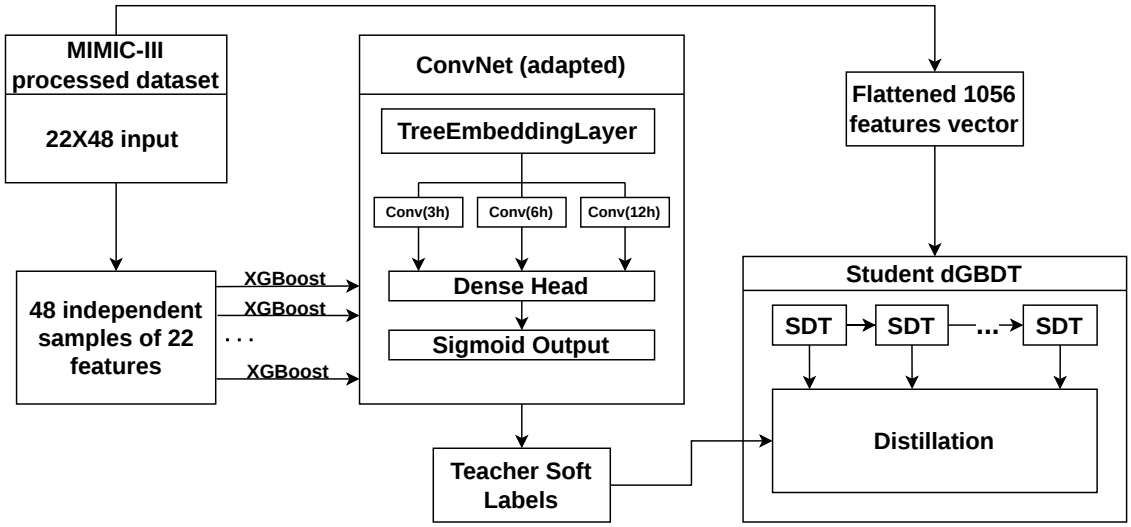


Figure 3.1: The TNT architecture adapted for the MIMIC-III dataset. The process flows from left to right: raw data is processed by XGBoost to generate leaf indices, which are then fed into a ConvNet teacher. The ConvNet's outputs (soft labels) and flattened inputs are used to train the dGBDT student model via distillation.

neural network to process. On the other hand, the Conv-SDT model (a "N-T" pattern) obtained slightly lower results than the ConvNet(0.8465 compared to 0.8484). This is not unexpected, as the SDT model itself is not so powerful. The result of the distilled soft tree sits right in-between the standalone SDT and the ConvNet. The TNT architectures both performed very well, proving that they managed to learn from the GBDT-Conv teacher. The distillation into a Soft Tree actually achieved the highest overall performance, peaking at an AUROC of 0.8822 and an AUPRC of 0.4818.

AUROC measures a model's separability power, and in this specific case it shows the probability that the model will rank a randomly chosen patient who died higher than one who survived. The problem is that AUROC can be too optimistic on imbalanced datasets. This is because the False Positive Rate is calculated as $FP/(FP + TN)$. In this ICU dataset, the number of True Negatives (TN) (correctly predicted survivors) is huge. A model can achieve a low False Positive Rate (and thus a high AUROC) simply by being very good at identifying the huge number of survivors, even if it performs poorly at identifying the few patients who die. This is why we measure AUPRC, which is much more sensitive to improvements for the positive class. The metric focuses on correctly identifying patients who are at risk of death, by measuring how well the model balances precision (ensuring the patients it flags are actually in danger) with recall (ensuring it doesn't miss any patients who are truly at risk).

For all described experiments, the training loop had 100 epochs with early stopping with patience 10, and the best model was restored from the previous epochs looking for

highest AUPRC, considering it is the most important metric when looking for the mortality risk. dGBDT models consist of 6 trees of depth 3, while the SDT models have depth 10 in the standalone version and 5 in Tree-Network variations. All distillation experiments were performed with an α value between 0.5 and 0.8 and distillation temperature equal to 1.

Methods		Imputed	
		AUROC	AUPRC
T	CART	0.7884	0.2954
	dGBDT	0.7445	0.2378
	SDT	0.8305	0.4107
N	Conv	0.8484	0.4312
T & N	GBDT-Conv	0.8739	0.4796
	Conv-SDT	0.8465	0.4159
TNT	GBDT-Conv-SDT	0.8822	0.4818
	GBDT-Conv-dGBDT	0.8732	0.4794

Table 3.2: Results for the MIMIC-III dataset with different models

Performing the imputation described by the ISeeU paper[3] removes about 80% of the missing values, but for patients with zero observations for a feature along the 48 hours, neither the forward nor backward fill can create them. Aligned with the method the same paper proposed, I use mean imputation for the remaining missing values for the teacher model and this behavior was persistent for all experiments described so far. However, adding the XGBoost part at the beginning of the model removes the necessity of imputing the data, as both the teacher and the student can handle missing values and use them in training. I decided to run 2 more sets of experiments where I don't use mean imputation: one where the data is not imputed at all and one where I keep the forward and backward imputation methods described by the ConvNet authors. The results from Table 3.3 show that relying on the tree-based models to handle the sparse data, increased the teacher model's performance. We can also see that the student models had a significant drop in both AUROC and AUPRC for the fully sparse data.

Based on another study[13] that performed knowledge distillation on MIMIC-III ICU mortality risk prediction task, I decided to try to aggregate the temporal features using summary statistics. The 48 hours of data from each patient is replaced by only 4 representative features: min, max, mean and variance. There are 16 temporal features and 6 static ones that were broadcasted for each hour to obtain the 22×48 bidimensional input. Aggregation reduced each sample from 22×48 to a unidimensional vector that has only

Model	Mean AUROC	Mean AUPRC
ffill and bfill		
GBDT-Conv	0.8740	0.4820
GBDT-Conv-SDT	0.8767	0.4800
GBDT-Conv-dGBDT	0.8754	0.4758
no imputation		
GBDT-Conv	0.8748	0.4880
GBDT-Conv-SDT	0.8581	0.4421
GBDT-Conv-dGBDT	0.8648	0.4497

Table 3.3: Results for the MIMIC-III dataset for different imputation strategies

$6 + 16 \times 4 = 70$ features. The teacher remains the same, trained on the temporal data, while only the student input dimensionality is reduced. The variations of this experiment include using the imputed dataset from the first part, the fully sparse one, and the middle ground where only `ffill` and `bfill` were applied along the temporal features of the same patient. The results from Table 3.4 show that even when the input space was compressed, the student networks managed to learn from the complex teacher ConvNet. The best results come from the dGBDT with patient-level imputation and the SDT with fully sparse data. The dGBDT architecture remained the same, while the SDT depth was increased to 7.

Model	Imputation	AUROC	AUPRC
SDT_AGG	Imputed	0.8652	0.4468
	ffill and bfill	0.8691	0.4592
	Sparse	0.8717	0.4781
dGBDT_AGG	Imputed	0.8698	0.4621
	ffill and bfill	0.8714	0.4668
	Sparse	0.8660	0.4538

Table 3.4: Results for the MIMIC-III dataset with aggregated data for the student models and different imputation strategies

Chapter 4

Interpretability

Moving to the most important part of distilling a deep neural network into a tree model, the inherent interpretability, I selected the student networks that showed best results to perform global and local analyses for a deep understanding of the model.

Global interpretability means looking at the model's behavior as a whole, aiming to make the entire decision-making logic transparent and understandable to humans. It allows you to see how a model weights different features across an entire dataset and to map out the rules and patterns it learned.

On the other hand, **local interpretability** means to move the focus to a single, specific prediction and to explain the reasoning behind the model's output for an individual sample[13].

Inherent interpretability is the property of a model of being interpretable by how it's built, without the need of any approximation tools. Tree models are inherently interpretable because you can look directly at every single node and see exactly what features were used to route the data on one branch or another. By contrast, neural networks miss this property because looking at any node that is not in the first layer will not be human-readable as the input is already altered by the previous layers.

4.1 Soft tree on temporal data

The best overall performance between all distillation experiments was achieved by the Soft Decision Tree trained on the full temporal data. The teacher is the ConvNet trained on patient-level imputed data, where the features that were fully missing from a patient were also mean-imputed with values from across the other patients, while the SDT used only the patient-level imputation and treated the fully missing data as extra information using the missingness mask. While the depth of the tree is only 5, resulting in only 31 internal nodes, the weight space of every single node is quite big. Still, we could look into the whole tree if needed. Looking inside a single node is really straightforward, we just

plot all weights in a bidimensional space, where on `y` axis we have the 22 features and on the `x` axis we have a temporal scale.

Because the data is patient-level imputed, for any single feature, there are only 2 possibilities: if the patient has at least one measurement for that feature, then the entire 48-hour vector will be populated with imputed data, or if the patient has zero measurements for that feature, the vector will be full of `NaN`. In the first case the missingness mask will be a constant vector of zeros, in the second case will be a constant vector of ones. Because of this, instead of multiplying 48 different mask values with the temporal weights, the equation can be simplified like this:

$$\sum_{t=1}^{48}(w_{mask_t} \times m_t) = m \times \sum_{t=1}^{48} w_{mask_t}$$

This means the model is not looking at each hour's mask at any point, it looks at the sum of all of them, so we can sum all the weights from the mask for a feature and treat it as the full missingness impact. This is exactly what I did in the plot from Figure 4.1.

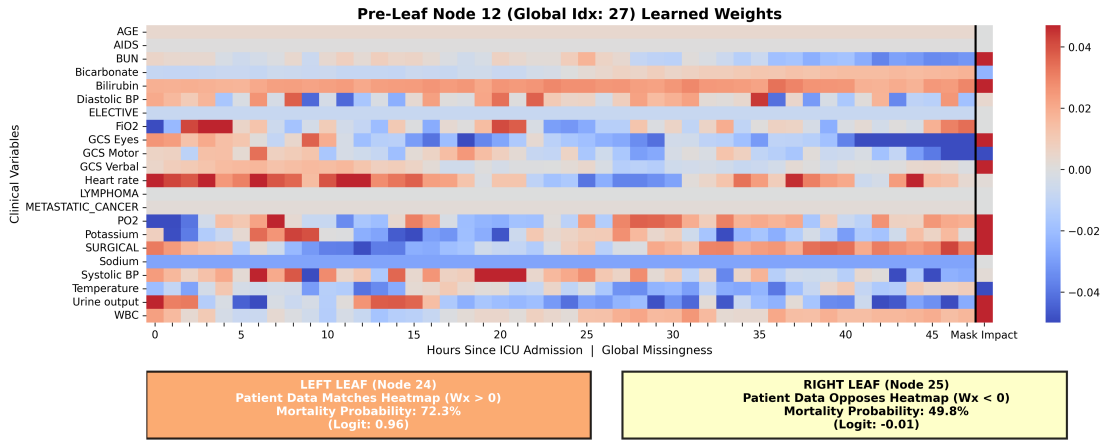


Figure 4.1: Heatmap of node 12 from SDT student.

The heatmap does not show higher or lower mortality risk, it shows a higher (red) or lower (blue) probability of going to the left. I selected node 12 to plot because its split goes to leaf nodes, so it is easier to actually see what going left or right would mean.

We can see in this plot that patients that arrive here can go either to the left leaf (72.3% risk) or the right leaf (49.9% risk). It is worth reminding that the patient will not actually have either 72.3% or 49.9% risk, because in a soft tree the data flows across all branches with different probabilities. These are the hypothetical probabilities if the node would route the patient 100% to the left or to the right.

Looking closely at the heatmap, a few major clinical patterns stand out that define this specific routing decision:

Late-Stage Neurological Deterioration revealed by the cluster of deep blue, negative

weights for GCS Eyes and GCS Motor during the final 8 hours of the observation window (hours 40–47). Because these weights are strongly negative, a lower actual GCS score (indicating unresponsiveness or coma) multiplied by this negative weight results in a large positive contribution to the node's logit. The model has learned that neurological decline right before the 48-hour mark is a massive red flag, pushing the routing probability heavily to the left. Conversely, a patient who maintains a normal, high GCS score will generate a negative mathematical contribution, pushing them toward the lower-risk right leaf (data is normalized with 0-mean, so low values are negative and highs are positive).

We can also observe positive (red) weights for Heart Rate across almost the entire 48 hours, and patches of strong positive weights for FiO2 (fraction of inspired oxygen), particularly in the early hours (2-5) and late hours (45-47). High, sustained heart rates and the need for elevated supplemental oxygen are classic signs of systemic stress, sepsis, or respiratory failure. Because these weights are positive, high actual measurements for Heart Rate and FiO2 directly add to the logit, pushing the patient toward the higher mortality leaf.

Looking at the "Mask Impact" column on the far right, we see the true value of treating missingness as a feature. There are intense dark red (positive) weights for missing PO2, Potassium, WBC, and Urine output. If a patient is missing these routine laboratory and fluid checks across their stay, the constant binary mask value is 1. Multiplying this 1 by the heavy positive weight adds a massive penalty to the patient's score, driving them into the 72.3% mortality leaf. This means that the specific cohort that arrives at this leaf has higher mortality risk if they are in a situation that does not require the lab test. This could reflect transitions to palliative care where routine checks are suspended. However, we can not know for sure without looking at the tree as a whole, to see exactly what types of patients arrive at this node with high probability. The tree is still too complex to be easy to interpret at this point, so moving to aggregated data for the student network might help for a better understanding.

4.2 Soft Tree on aggregated data

Looking at each feature by itself in the temporal setting can still be difficult to follow. Because the input space is so big, it could be confusing for someone to look at hour 26 and hour 32 measurements of heart rate because the model said that those are the most important features. It is easier to interpret that a big heart rate mean or variance is a sign for a higher mortality risk for example.

Looking at the whole tree might still be too much so we need to compute specific interpretation components to make it easier to understand what is happening inside the tree. This is why I extracted the global feature importance for this model. Global importance provides a macroscopic view of the model's logic, revealing exactly which

clinical measurements drive the predictions.

To quantify this, I defined global importance as the expected weight contribution of each feature across all internal routing nodes. For a given feature f, the importance is calculated as the sum of its absolute weights $|W_{j,f}|$ at each node j, multiplied by the mean probability μ_j of a patient reaching that node during the validation phase. We can look at the results in Figure 4.2 and see that the most striking observation is the model's heavy reliance on the missingness mask. The highest single importance score in the entire matrix belongs to the missingness of GCSEyes (0.039), followed closely by GCSMotor missingness (0.027) and FiO2 missingness (0.024). This strongly aligns with the behavior observed in the temporal tree: the absence of data is often just as informative as the data itself. In order to understand why this could happen we need to have a better look at the data, especially at the patients that miss at least one of the Glasgow Coma Score measurements. I added the analysis results to Table 4.1.

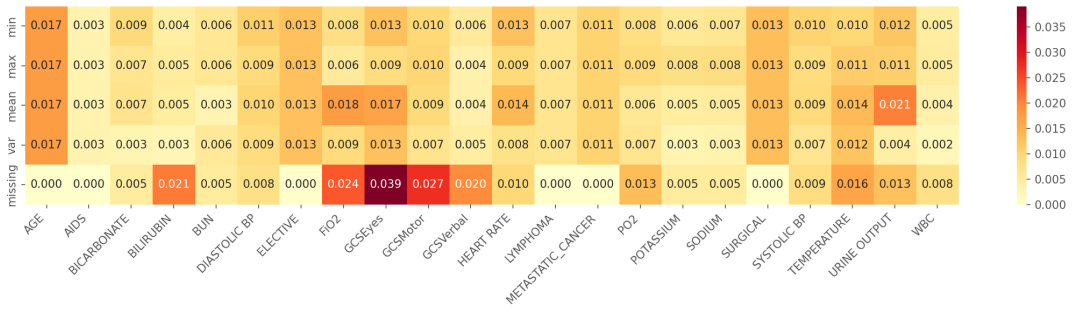


Figure 4.2: Global Feature Importance for SDT on aggregated data

I also extracted the following statistics about the missingness of the Glasgow Coma Score components:

```
================================================================================
 GCS COMPONENT MISSINGNESS & OVERLAP METRICS
================================================================================
Total Cohort Size:                 20372 patients
Missing None (Complete GCS):       3268 patients (16.04%)
--- Individual Missingness (Inclusive) ---
Missing Eyes:                      918 patients (4.51%)
Missing Verbal:                    14862 patients (72.95%)
Missing Motor:                     10639 patients (52.22%)
--- Pairwise Missingness (Inclusive of 3rd component) ---
Missing Eyes & Verbal:             904 patients
Missing Eyes & Motor:              445 patients
Missing Verbal & Motor:            8409 patients
--- Missing All Three Components ---
Missing All Three:                 443 patients (2.17%)
================================================================================
```

	Missing GCS Eyes		Missing GCS Motor		Missing GCS Verbal	
Feature	**Missing**	**Has**	**Missing**	**Has**	**Missing**	**Has**
Continuous Features (Mean)						
AGE	59.40	63.84	63.80	63.47	62.41	66.96
BICARBONATE	22.38	23.84	24.18	23.37	23.78	23.81
BILIRUBIN	1.83	2.00	1.93	2.03	1.84	2.38
BUN	23.81	25.22	24.66	25.70	24.56	26.76
DIASTOLIC BP	62.84	60.30	59.77	61.05	59.99	61.45
FiO2	45.17	39.69	35.39	44.90	40.90	37.32
GCSEyes	NaN	3.53	3.84	3.18	3.53	3.52
GCSMotor	3.78	4.64	NaN	4.60	4.53	4.74
GCSVerbal	2.69	3.62	3.88	3.44	NaN	3.62
HEART RATE	88.82	86.40	85.65	87.36	86.46	86.53
PO2	140.92	125.90	119.65	131.64	128.15	122.10
POTASSIUM	4.05	4.10	4.12	4.07	4.11	4.05
SODIUM	140.71	138.65	138.11	139.39	138.56	139.17
SYSTOLIC BP	121.93	120.12	118.90	121.55	118.50	124.63
TEMPERATURE	37.15	36.94	36.84	37.07	36.98	36.87
URINE OUTPUT	132.84	142.02	163.62	118.42	147.60	126.08
WBC	13.50	12.40	12.04	12.88	12.48	12.33
Static/Binary Features (% Positive)						
AIDS	0.9	1.0	1.0	0.9	1.1	0.7
ELECTIVE	8.2	14.7	16.7	11.9	15.8	10.5
LYMPHOMA	1.7	2.9	3.1	2.6	2.8	2.9
METASTATIC_CANCER	2.7	6.1	6.7	5.2	5.9	6.2
SURGICAL	38.8	44.9	46.9	42.2	46.6	39.3

Table 4.1: Characteristics of Patients by Missing GCS Components

Missing GCSVerbal (The Intubated Patient): These patients are generally younger than those with complete records and have higher rates of Surgical and Elective admissions. Based on the increased FiO2 values, they require supplemental oxygen. The explanation could be that the patients have an endotracheal tube in place, making speech physically impossible. Clinicians often chart this as "1T" (Tubed), which data pipelines convert to a missing value.

Missing GCSEyes (Severe Trauma or Deep Coma): This is the rarest type in the cohort and the most severe. Almost all of them also miss GCSVerbal, but they have highest FiO2 requirements and lower overall Glasgow coma scores. The elective and surgical rates are also lower. An inability to assess eye-opening is very rare, and it could happen for two reasons: severe facial trauma (eyes are swollen shut) or profound coma / deep pharmacological sedation.

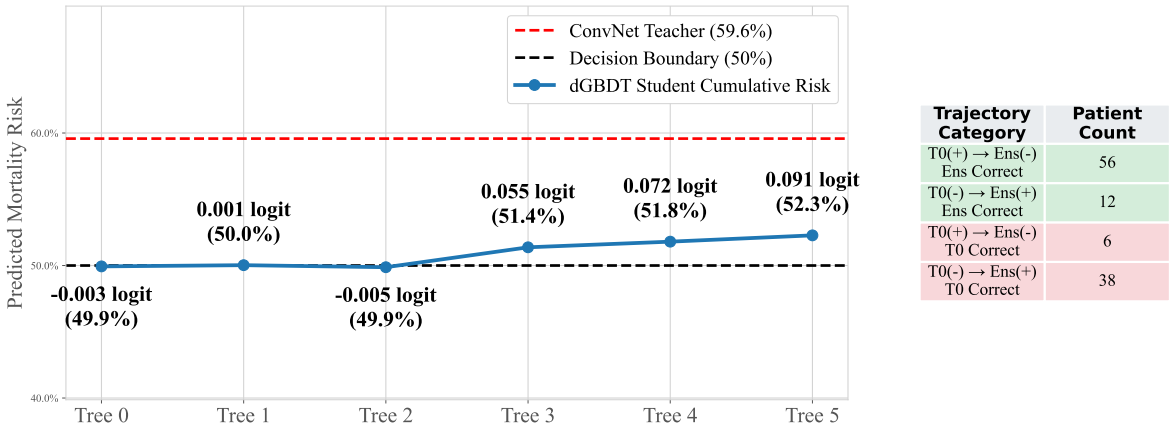
Missing GCSMotor (Stable or Regional Anesthesia): Paradoxically, patients missing their Motor score have the highest scores in their remaining GCS components. Their GCS Eyes mean is 3.84 and GCS Verbal is 3.88 (approaching the maximum of 4). They have the highest rates of elective and surgical admissions, and much lower FiO2 requirements than the other categories. These are probably patients recovering from regional anesthesia or epidural, where a standard GCS motor assessment is temporarily impossible.

These profiles would explain why the missing GCSEyes is the single most important feature for the model, and also why in the node from the SDT on temporal data we saw

that missing eyes and verbal measurements shifted the model's decision to the higher mortality risk leaf, while missing the motor score shifted to the leaf with lower mortality risk.

4.3 dGBDT on aggregated data

The boosting model has the advantage of using shallow trees for making predictions, in this case the depth is only 3. This means we can look at a whole tree at once and understand what is happening behind the scenes. This makes this type of model highly suitable for local interpretations, making it easy to track a patient's data journey across the ensemble. For example, we can check how each tree contributes to the final prediction for a specific patient's risk. For the visualization in Figure 4.3 I selected a patient that can highlight the power of the ensemble model, because the first tree actually makes a prediction in the wrong direction, then the following trees make a correction. It is true that this is not always the case, if we check all patients from the validation set we find that for a total of 68 cases the ensemble changing the prediction led to an improvement, while in 44 cases **tree 0** was right and the rest of the trees changed the prediction to wrong. The most valuable are the 12 cases where the last tree's correction led to finding patients that actually were in the mortality class.



Trajectory Category	Patient Count
T0(+) → Ens(-) Ens Correct	56
T0(-) → Ens(+) Ens Correct	12
T0(+) → Ens(-) T0 Correct	6
T0(-) → Ens(+) T0 Correct	38

Figure 4.3: Trajectory of patient with ID 44514 in the dGBDT ensemble

Seeing that the **tree 3** made the most important correction makes us want to look deeper inside and see exactly what made it take that turn. In Figure 4.4 we can see the patient's trajectory across the tree. The nodes are represented as 22×5 heatmaps with each feature's contribution, that is computed by multiplying the node's learned weight with the sample's feature value ($W_i \times X_i$). At each internal node, the sum of those feature contributions (plus a bias term) is passed through a sigmoid activation function. This outputs a value between 0 and 1, which dictates the probability of the patient routing to

the left child branch ($\sigma(W \cdot x + b)$) and the right child branch ($1 - \sigma(W \cdot x + b)$). Because this is a soft decision tree, the patient doesn't take a single hard path. They begin at the root with 100% mass. At every split, that mass is multiplied by the left and right routing probabilities, until we get the final mass contribution of each leaf.

In the heatmap, blue represents negative and red positive, but because the data was centered around 0, we should think that blue means low and red means high, and the intensity is how far from the mean the value is. For binary features even if the color is not that intense, blue represents 0 and red 1.

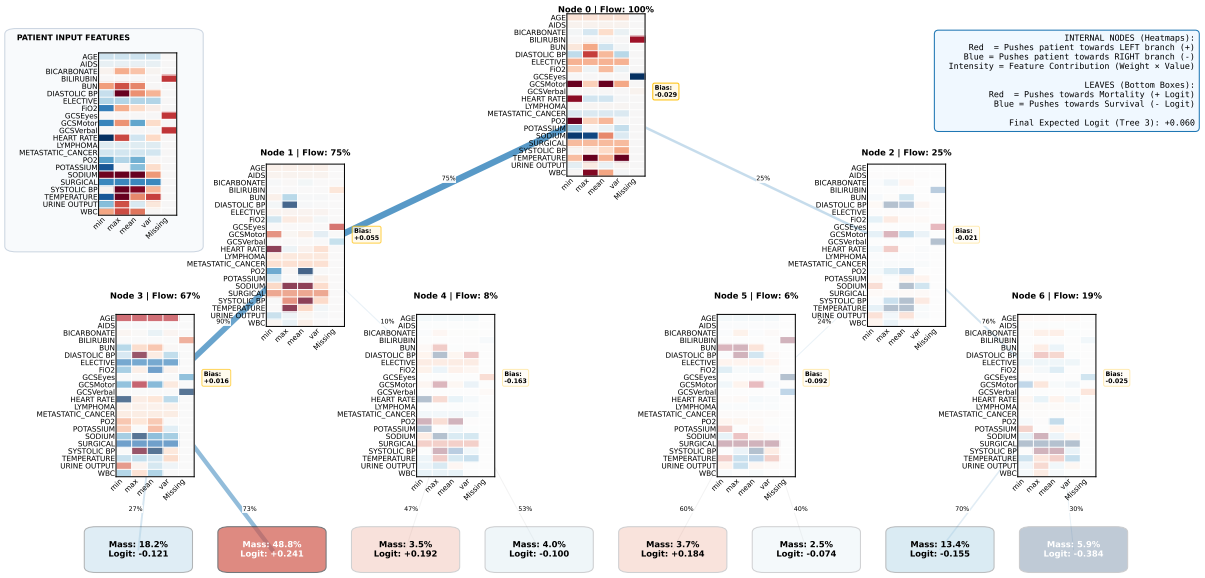


Figure 4.4: Routing of patient 44514 through Tree 3 of the dGBDT ensemble. Each internal node is a 22×5 contribution heatmap ($W_i \times X_i$). Edge thickness scales with probability mass. The dominant path carries 48.8% of the patient's mass to the leaf with the highest positive logit, making it the primary driver of this tree's correction toward mortality.

Before looking at the nodes we should first see the patient's data. The heatmap with the patient features provides immediate insights into the patient's profile before any individual nodes are examined. In the missingness column we see values for GCSEyes, GCSVerbal, and Bilirubin, meaning none of these were recorded across the entire 48-hour window. Based on the profiles discussed earlier, the combination of missing eye and verbal assessments and the available motor scores are a strong indicator of a sedated or intubated patient. The sodium rows are red, indicating persistently elevated sodium levels well above the cohort mean. The GCSMotor has low values, particularly in the minimum statistic, which signals neurological compromise at some point during the stay. Blood pressure statistics have high values, while PO2 sits clearly below average across all statistics. The minimum heart rate is also very low.

At the root **Node 0**, we see the first split leans 75% left, based mostly on the minimum values of GCSMotor, heart rate and PO2 and high maximum and variance for Temperature, showing the critical condition of the patient at some point during the stay. Missingness in GCSEyes actually shifts the decision a little to the right. Here, the tree is detecting acute physiological instability. Low PO2 and low heart rate minima combined with a high and variable temperature point to a patient who experienced a severe episode during the stay.

Node 1 receives 75% of the mass and issues the most confident split in the entire tree, with a left probability of 90%. It still looks at some of the features from before, but also takes into account that the patient is not surgical, does not have lymphoma or metastatic cancer, does not have AIDS and is not an elective patient. The acute physiological signals from Node 0 already flag instability, but the absence of surgical admission, elective status, and chronic oncological or immunological conditions narrows the profile to spontaneous emergency.

Node 3 is the last split before the leaves and the last node that has over 50 percent of the flow. The low minimum heart rate, high maximum sodium, the fact that it is not elective and not surgical shift the decision to the right, alongside with the missingness from the GCS Verbal and Motor. Hypernatremia reaching a high peak is a marker of severe dehydration or impaired renal water handling, both of which are associated with poor outcomes in critically ill patients. Combined with bradycardia at some point during the stay and the absence of any neurological verbal or motor assessments, this node may be characterizing a patient in late-stage deterioration or even a severe burn case where standard neurological monitoring has become impractical.

The leaf with highest logit value has also the highest mass, with almost 50% of the decision of the tree being dictated by it, so the patient profile matches this tree's routing path for highest mortality risk.

Chapter 5

Conclusions

A primary achievement of this study is demonstrating that the student models, a distillable Gradient Boosting Decision Tree (dGBDT) composed of Soft Decision Trees (SDT) and SDT alone, successfully preserved the high predictive performance of the ISeeU ConvNet teacher. Evaluated on the MIMIC-III database for ICU mortality risk prediction, the proposed distillation framework successfully transferred the complex "dark knowledge" embedded within the ConvNet teacher. The distilled tree models proved to be highly competitive with the neural network, showing that performance does not need to be sacrificed to achieve model transparency. Even more, the SDT student obtained slightly better results (AUROC 0.8822 and AUPRC 0.4818) than the teacher network (AUROC 0.8739 and AUPRC 0.4796).

Additionally, this thesis introduced a specialized masking mechanism to handle the sparse nature of clinical temporal sequences. By treating missing observations as distinct features rather than using imputation to cover them, the model learned how to use missing data in its favor. Global interpretability analysis showed that the absence of data, such as missing components of the Glasgow Coma Score or routine laboratory tests, could carry deep clinical significance, that the models used to draw high mortality risk patient profiles and make predictions according to them.

Finally, the differentiable tree architectures unlock the reasoning mechanisms of deep learning. Through global and local interpretability techniques, medical professionals can understand the model's decision-making process and track a specific patient's trajectory across the ensemble, visualizing exact feature contributions and probability flows at every internal node. By successfully bringing together the advantages of powerful neural networks and transparent tree-based structures, this work showcased trustworthy, high-performing machine learning models that could be used in critical care medicine and other high-stakes fields.

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